



Deep learning-based automatic cranial implant design through direct defect shape prediction and its comparison study

Afaque Rafique Memon^{1,2,4} · Haochen Shi¹ · Tarique Rafique Memon⁵ · Jan Egger^{3,4} · Xiaojun Chen¹ 

Received: 11 December 2024 / Accepted: 9 April 2025 / Published online: 2 May 2025
© International Federation for Medical and Biological Engineering 2025

Abstract

Defects to human crania are one kind of head bone damages, and cranial implants can be used to repair the defected crania. The automation of the implant design process is crucial in reducing the corresponding therapy time. Taking the cranial implant design problem as a special kind of shape completion task, an automatic cranial implant design workflow is proposed, which consists of a deep neural network for the direct shape prediction of the missing part of the defective cranium and conventional post-processing steps to refine the automatically generated implant. To evaluate the proposed workflow, we employ cross-validation and report an average Dice Similarity Score and boundary Dice Similarity Score of 0.81 and 0.81, respectively. We also measure the surface distance error using the 95th quantile of the Hausdorff Distance, which yields an average of 3.01 mm. Comparison with the manual cranial implant design procedure also revealed the convenience of the proposed workflow. In addition, a plugin is developed for 3D Slicer, which implements the proposed automatic cranial implant design workflow and can facilitate the end-users.

Keywords Deep learning · 3D medical imaging · Automatic implant design · Defect shape prediction · Implant customization · Medical AI algorithms

1 Introduction

In current medical practice, the design and manufacturing of cranial implants are typically handled by external suppliers, requiring costly proprietary software and highly trained designers [1, 2]. The three-dimensional reconstruction of

defective craniofacial regions, particularly in cases of large and complex defects, is a challenging task. It demands sophisticated image processing algorithms and advanced computer-aided design (CAD) tools. Consequently, the process remains time-intensive and expensive. Automatic cranial implant design has thus become a key area of research, with numerous studies seeking to develop algorithms that assist doctors in generating cranial implants directly from CT scan data [3].

Cranial implants are essential in cases of severe cranial defects caused by trauma, tumors, or congenital anomalies [4]. Each year, an estimated 0.7 million cranial repair operations are performed globally [5]. While patient-specific implants (PSI) are not always mandatory, they are often preferred for large defects, as alternative methods may fail to provide optimal aesthetic and functional outcomes [6]. The PSI design process can be framed as a shape-completion problem, where the goal is to restore the cranial structure's natural symmetry and appearance.

Traditionally, PSI design is outsourced to third-party suppliers, resulting in significant delays [7]. With advancements in additive manufacturing, such as 3D printing, it has become feasible to design and fabricate implants within

Afaque Rafique Memon and Haochen Shi contributed equally to this work.

✉ Xiaojun Chen
xiaojunchen@sjtu.edu.cn

¹ School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai, China

² Department of Mechanical Engineering, Mehran University of Engineering and Technology, Jamshoro, Sindh, Pakistan

³ Institute of Computer Graphics and Vision, Graz University of Technology, Graz, Austria

⁴ Institute for AI in Medicine, University Hospital Essen, Essen, Germany

⁵ Department of Electronic Engineering, Quaid-e-Awam University of Engineering, Science and Technology, Nawabshah, Sindh, Pakistan

hospital settings. This shift could drastically reduce patient waiting times. However, cranial implant design is a complex task that typically falls under the expertise of engineers rather than surgeons, posing a challenge for on-site implementation.

Recent advancements in artificial intelligence (AI) and deep learning have opened new avenues for automating cranial implant design. AI-driven methods address many limitations of manual design processes, particularly their dependence on time, expertise, and symmetry-based approaches. Early innovations, such as the convolutional neural network (CNN)-based method proposed by Li et al., introduced a coarse-to-fine framework for generating cranial implants directly from defective cranium data [8]. This approach reduced manual intervention and accelerated the design process. Subsequent studies refined these methods by employing patch-based training and inference strategies to handle high-resolution cranial images, further enhancing precision and reliability [9].

Other researchers have proposed an alternative workflow involving complete cranium reconstruction as an intermediate step before implant generation. Wu et al. developed a comprehensive 3D deep learning system that generated training datasets for diverse cranial defect scenarios and successfully applied the system in clinical surgeries [10]. Similarly, Mainprize et al. presented a U-Net-based pipeline to predict complete cranium models and derive implants from them, offering greater flexibility for complex cases [11]. Bayat et al. adopted a hybrid approach that first reconstructed defective crania in low-resolution 3D before refining them in high-resolution 2D slices, yielding favorable outcomes for cranial restoration [12].

Despite these advancements, existing approaches face challenges when working with high-resolution CT images. Many methods rely on segmenting images into smaller regions to accommodate computational constraints, often leading to discontinuities at the segment junctions. These discontinuities compromise the implants' structural and aesthetic accuracy. To overcome this limitation, we propose an automatic cranial implant design workflow that processes entire head CT images without segmentation. By employing a unified deep neural network architecture, our method ensures seamless

and continuous implant generation, addressing the limitations of existing approaches.

To validate the proposed method, we conducted comprehensive experiments using real head CT scans, evaluating its accuracy and clinical viability across diverse patient cases. Additionally, our approach has been integrated into 3D Slicer, an open-source medical image processing platform widely used in clinical and research settings [13]. This integration provides clinicians with a user-friendly, efficient tool to streamline cranial implant design, reducing reliance on external suppliers and enhancing patient care.

By leveraging deep learning technology and addressing challenges such as segmentation discontinuity, our research aims to advance cranial implant design. Our solution represents a significant step toward improving neurosurgical outcomes and the quality of life for patients requiring cranial restorative interventions.

2 Methodology

The proposed method involves predicting the defect part of a resized defective cranium and adopting several post-processing steps to refine the prediction toward a feasible implant. As illustrated in Fig. 1, the proposed approach employs a 3D inpainting (completion) network with input and output resolutions set to $128 \times 128 \times 128$ voxels. To apply this network, the bone part of a defective cranium image, along with necessary margins, must first be resized to match this resolution. The proposed network can be used to predict the missing part of the defective cranium, and then the resulting is subjected to connectivity analysis, inverse resizing, and Gaussian blur to obtain the final cranial implant.

2.1 Structure

The proposed deep neural network has a U-Net [14] like structure, with residual convolutional blocks [15] in the encoder part and ordinary convolutional blocks in the decoder part, as shown in Fig. 2. The input and output of the network are both volumes in the resolution of $128 \times 128 \times 128$ voxels. The number of filters is initially set

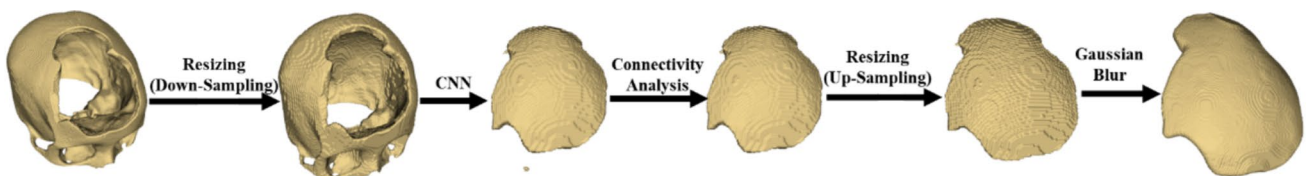


Fig. 1 The proposed workflow for automatic cranial implant design

Fig. 2 Structure of the proposed defect shape prediction network. The number of filters for each convolutional layer is denoted in the parentheses

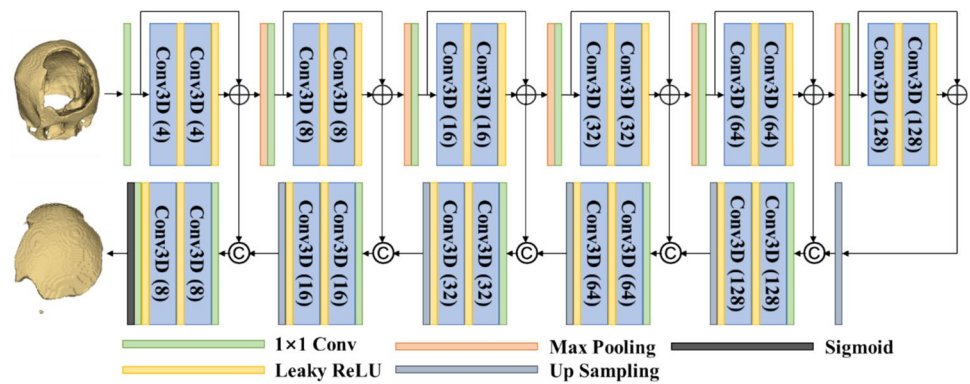
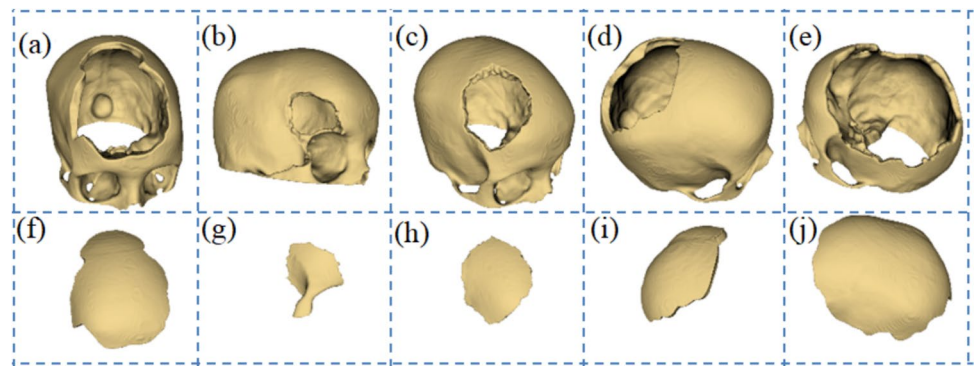


Fig. 3 Sample data taken from the dataset for task 1 of the AutoImplant 2021 Challenge (<https://autoimplant2021.grand-challenge.org/>). **a–e** Defective crania with defects at biface (a), frontal orbit (b), frontotemporal (c), and two other random positions (d, e); **f–j** corresponding ground truth implants for **a–e**, respectively



to four and will be doubled with each decrease of resolution. Kernel size of each 3D convolutional layer is set to the same value $3 \times 3 \times 3$. Max pooling is used to halve the volume resolution in each axis, while up-sampling is used to double the volume resolution. Leaky ReLU [16] is used as activation function for most layers, except the final output layer, in which a sigmoid function is chosen to regulate the output voxel values to the range between zero and one. In the bottleneck of the network, the volume resolution is only $4 \times 4 \times 4$, and the number of filters is 128. The total number of trainable parameters of the proposed network is approximately 2.4 million, which is quite lightweight compared to other researchers' networks [17].

2.2 Dataset

The training and testing are mainly conducted on the dataset taken from the task 1 of the openly accessible AutoImplant 2021 Challenge [18]. This dataset contains 114 health human crania, and five synthetic defective crania were created for each health cranium with defects at biface, frontal orbit, top temporal and two other random positions, respectively, as shown in Fig. 3. Thus, there are a total of 570

Table 1 Partitioning of the training, validation, and testing sets for each fold

	Training set	Validation set	Testing set
Fold #1	0 ~ 66	66 ~ 75	76 ~ 113
Fold #2	0 ~ 38	104 ~ 113	38 ~ 75
Fold #3	38 ~ 103	104 ~ 113	0 ~ 37

cases in the dataset, and for each case, the defective and complete crania, as well as the corresponding implant, are provided. These models are stored as binary voxel volumes in the nrrd format, with value one representing the bone and value zero representing the background. The resolution of each volume is $512 \times 512 \times Z$ voxels, where Z varies between different cases.

2.3 Training

The proposed network is implemented using Keras in a workstation with one RTX 6000 graphics card. Since threefold cross-validation is chosen to evaluate the performance of the proposed network, the 114 healthy human crania are randomly split into three groups, and for each

fold, 380 cases from two of the groups are used for training and validation, while the other 190 cases from the last group are used for testing. The detailed partitioning scheme for each fold is shown in Table 1. The well-known Dice loss is chosen as the loss function, and Adam optimizer with a learning rate of 0.0002 is used. For each fold, the network is randomly initialized and trained in the training set for 50 epochs.

2.4 Post-processing

The output of the proposed deep learning network is a probability graph indicating the probability of each pixel belonging to the implant. Firstly, the connectivity analysis is used to select the largest non-zero output region. Then, an inverse resizing is conducted to fit the predicted implant into the original defective cranium volume. After that, Gaussian blur is used to smooth the artifacts caused by up-sampling. Finally, the probability graph is converted into a binarized voxel image at the threshold of 0.5, which is the final cranial implant of the proposed workflow.

2.4.1 Connectivity analysis

On the assumption that one defective cranium will only have one defective position, and the commonly observed fact that deep learning network will sometimes produce some noise outside the implant region, in the proposed workflow, only the largest connected region of the network output is considered as the expected cranial implant. After removing the non-largest, non-zero output regions, the probability graph will have only one connected region, which is supposed to be the original shape of the missing region on the defective cranium.

2.4.2 Gaussian blur

Since the voxel resolution of the original defective cranium is higher than the voxel resolution during processing, jagged edges will occur during up-sampling. To make the edges of the final implant smooth, a Gaussian blur with a standard deviation of four voxels is applied to the up-sampled implant volume.

The definition of Gaussian kernel used in Gaussian blur is as follows:

$$G(x, y, z) = \frac{1}{(2\pi\sigma^2)^{3/2}} e^{-\frac{x^2+y^2+z^2}{2\sigma^2}}$$

where x, y, z denote for the voxel offset from the center, and σ denotes for the standard deviation of the Gaussian blur. Computationally, voxels 3σ away from the center contribute little to the result and thus negligible.

The up-sampled implant will be blurred by convolution with the Gaussian kernel, and then binarized again.

3 Evaluation metrics

To measure the performance of the proposed workflow, three metrics between the generated cranial implants and their corresponding ground truth implants are calculated, which are the Dice Similarity Score (DSC), boundary DSC (bDSC), and 95th quantile of Hausdorff Distance (95hd).

Using y_p and y_g to denote the automatic prediction result and its corresponding ground-truth, the DSC is calculated for each case as follows:

$$DSC = \frac{2|y_p \cap y_g|}{|y_p| + |y_g|}$$

where $y_p \cap y_g$ denotes for element-wise Boolean AND operation for y_p and y_g , and $|y_p|$ calculation the number of non-zero elements in y_p . If y_p and y_g are the same, the DSC of that case will equal 1. Otherwise, lower DSC means worse prediction result for the whole implant.

Like DSC, bDSC calculates the DSC only for the boundary part of the two implants. Using x_d to denote the input defective cranium, function $d(y; x_d)$ maps each voxel in the volume y to its minimum distance from the surface of x_d . Given a threshold distance D , and define a filter function F as follows:

$$F(y; x_d, D) = y \cdot \Phi[d(y; x_d) \leq D]$$

where $\Phi[X]$ equals to 1 when the Boolean condition X evaluates to true, otherwise 0. Since function F keeps only the boundary part of an implant, the bDSC can be expressed as follows:

$$bDSC = \frac{2|y'_p \cap y'_g|}{|y'_p| + |y'_g|}$$

where $y'_p = F(y_p; x_d, D)$ and $y'_g = F(y_g; x_d, D)$.

Given two surfaces, the Hausdorff distance measures the maximum value of the distance for each point in one surface to its closest point in the other surface. In other words, one surface needs to be moved at least the Hausdorff distance to cover any given point in the other surface completely. Because there may be some outliers in the generated implant, which can greatly impact the Hausdorff distance, the 95th quantile of Hausdorff distance (95hd) is used to give a more reasonable evaluation of the likelihood of two surface, and in our case, the surfaces of the automatically generated implant and its corresponding ground-truth.

4 Integration with 3D slicer

3D Slicer (Brigham and Women's Hospital, Harvard Medical School, USA) is a widely used open-source software maintained by its community, which implements many easy-to-use functionalities for the visualization, segmentation, and registration of medical images. 3D Slicer is also known as an excellent medical image processing platform for its outstanding extensibility. To facilitate the usage of end-users, a plugin for 3D Slicer (version 5.0.2) is developed by us to implement the proposed automatic cranial implant design workflow in the “one-click” manner. Once a specific remote server is started, as shown in Fig. 4, the user only needs to select the volume name of the defective cranium, set the output volume name in the 3D Slicer, and click the “Apply” button. Then, the plugin will send the volume image containing defective cranium to the remote server, and the server will automatically start the prediction workflow. After about 7s, a new volume containing the generated cranial implant will be added to 3D Slicer, and the user is allowed to conduct any further modifications where necessary.

5 Results

To evaluate the performance of the proposed workflow, three key metrics are calculated: Dice Similarity Score (DSC), 95th quantile of Hausdorff Distance (95hd), and

Table 2 Average evaluation metrics for each fold, regardless of defective positions

	DSC	bDSC	95hd (mm)
Fold #1	0.81 ± 0.08	0.82 ± 0.06	2.89 ± 1.75
Fold #2	0.83 ± 0.07	0.83 ± 0.06	2.94 ± 2.94
Fold #3	0.80 ± 0.10	0.79 ± 0.09	3.17 ± 2.48

boundary DSC (bDSC). The average evaluation metrics across all folds are as follows: DSC = 0.81, bDSC = 0.81, and 95hd = 3.01 mm. As shown in Table 2 and Fig. 5, there are no significant differences between the metrics across different cross-validation folds, indicating consistent performance of the model regardless of the defective positions.

Evaluation metrics are also calculated for each type of defective positions. As shown in Table 3 and Fig. 6, the outcomes with defects at the frontal orbit are significantly worse than other positions, which is assumed to be attributed to the complex and distinctive structure of the frontal orbit.

To ensure the clinical viability of our approach, it is essential to contextualize these metrics and assess their relevance in neurosurgical practice. The Dice Similarity Coefficient is a widely used metric for measuring the similarity between two sets of data ranging from 0 to 1, with high values indicating better overlap between the predicted and ground truth data. A DSC of 0.81 suggests

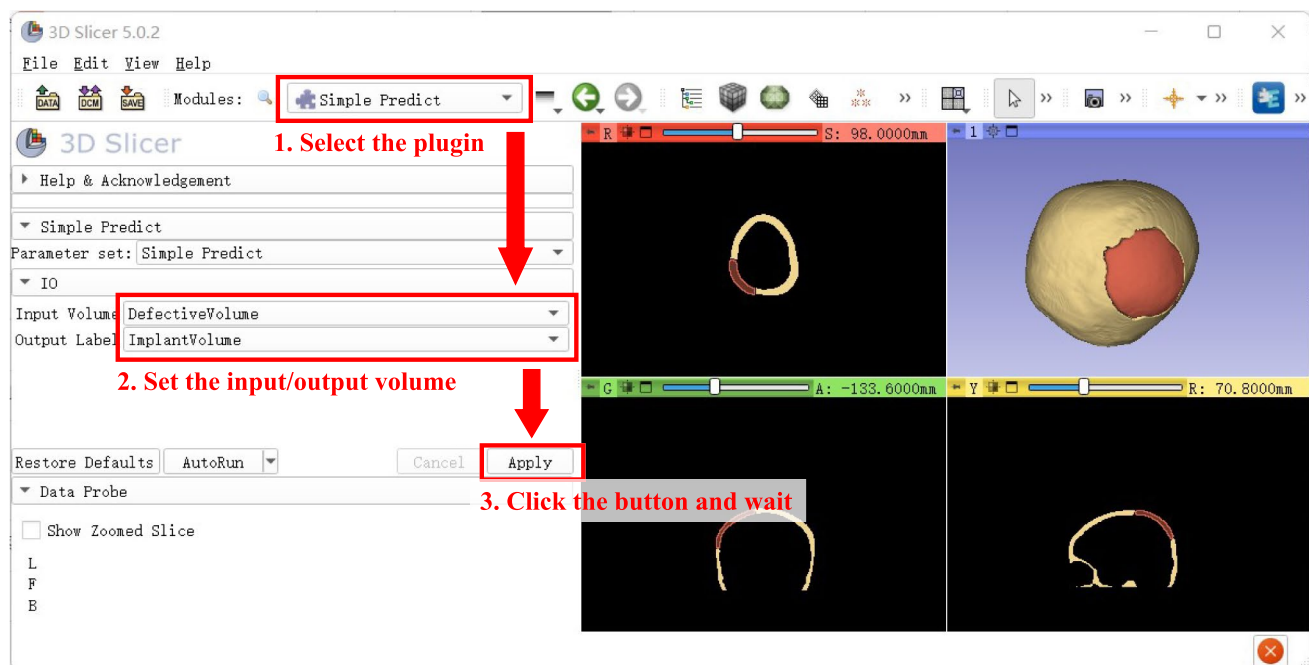


Fig. 4 Usage of the developed automatic cranial implant design plugin for 3D slicer

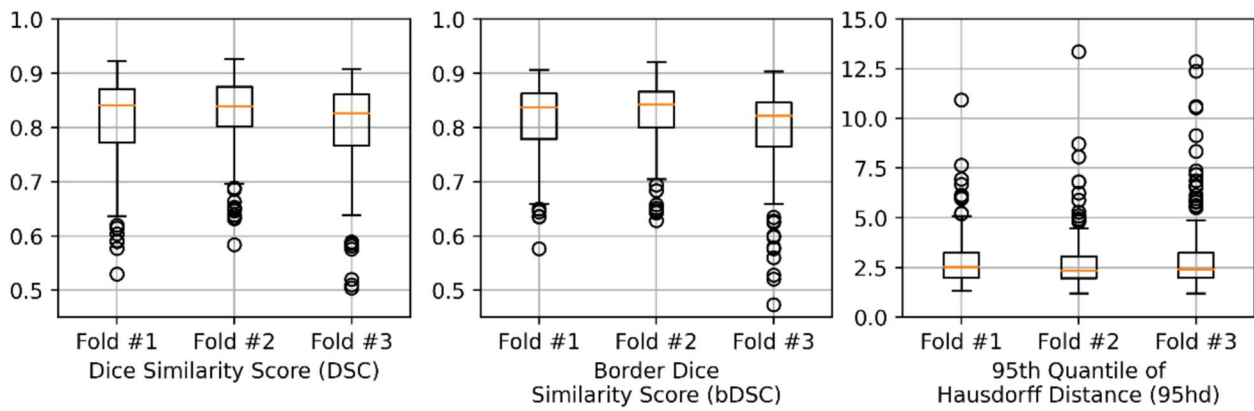


Fig. 5 Boxplot of the evaluation metrics for each fold, regardless of defective positions

Table 3 Evaluation metrics for different defective positions

	DSC	bDSC	95hd (mm)
Biface	0.84 ± 0.07	0.84 ± 0.05	2.64 ± 1.31
Frontal orbit	0.76 ± 0.09	0.75 ± 0.09	4.15 ± 3.02
Top temporal	0.83 ± 0.06	0.83 ± 0.05	2.32 ± 1.08
Random	0.82 ± 0.08	0.82 ± 0.06	2.97 ± 2.82

a substantial agreement between the generated cranial implants and the actual cranial defects, indicating a high degree of accuracy in our prediction [19]. The boundary DSC is an extension of the DSC that specifically evaluates the agreement at the boundaries of the predicted and ground truth data. A bDSC of 0.81 further confirms the consistency and precision of our method in accurately capturing the intricate details of the cranial defects, which is crucial for successful cranial implant and surgical outcomes [20].

Additionally, the 95th percentile Hausdorff distance (95hd) quantifies the maximum distance between the predicted and ground truth surfaces. In the context of cranial implant design, a 95hd value of 3 mm suggests that the

majority (95%) of the points on the surface of the predicted cranial implant are within a 3-mm margin from the corresponding points on the ground truth, demonstrating an impressive level of anatomical fidelity and alignment [21].

Besides the quantitative evaluation of the proposed workflow, some cases are also plotted for a qualitative evaluation as shown in Fig. 7. According to the dataset provider, all the defective crania are created from the corresponding complete crania by cutting off a part of the region, which means that the intersection between one defective cranium and its corresponding ground truth implant are none, and the union of them is exactly the original complete cranium. However, this also means that the provided ground truth implants will inevitably have some glitches near the edges, which can be seen obviously in the column b of Fig. 7. Since Gaussian blur is applied, the edges of the implants generated from the proposed workflow will be smoother than their corresponding ground truths, as shown in the column c of Fig. 7. Column d and e indicate that the automatically generated implants tend to be more upwardly convex than the corresponding ground truths, and the color legends also show that the

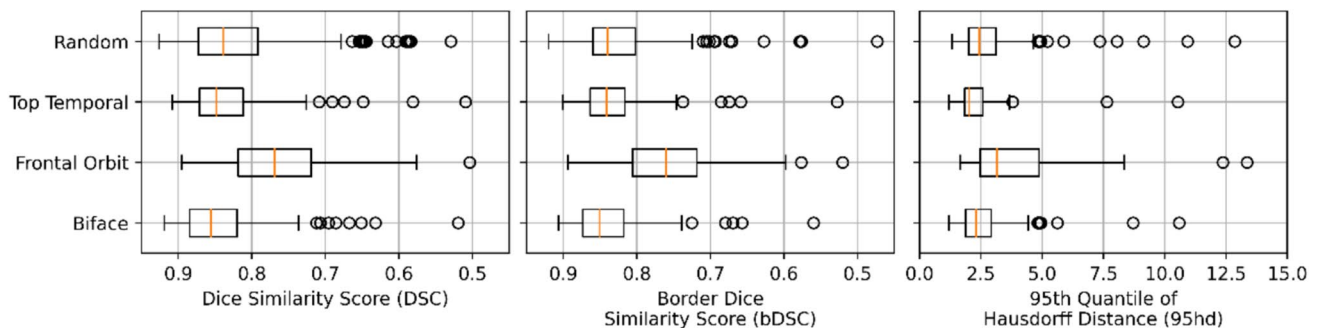
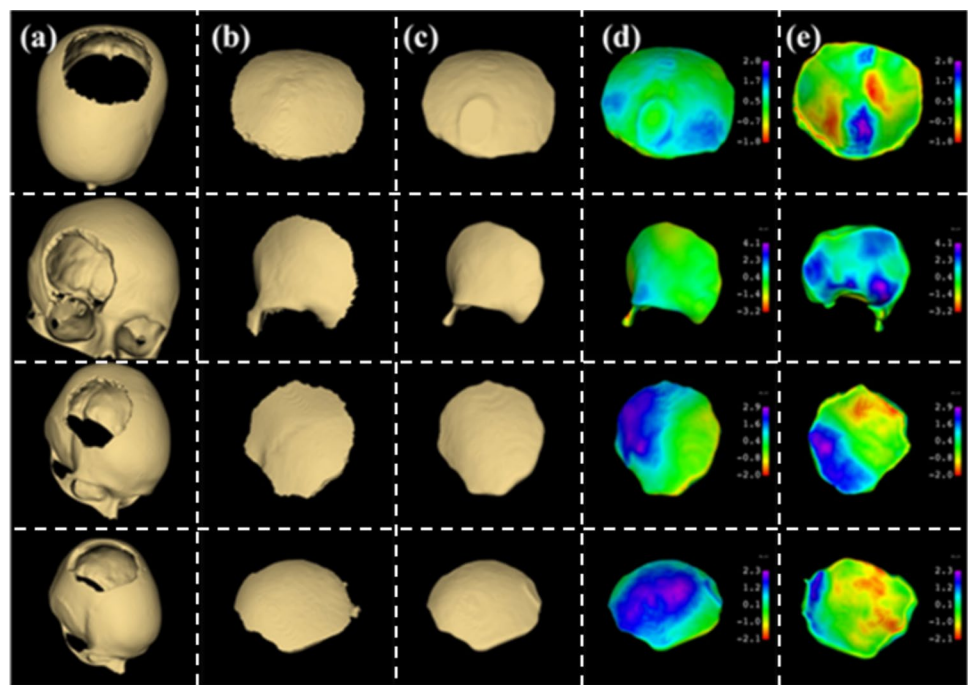


Fig. 6 Boxplot of the evaluation metrics for different defective positions

Fig. 7 Graphical views of the automatically generated cranial implants and their comparisons with the corresponding ground truths. From top to bottom, each row represents one defective case from the same healthy cranium with the synthesized defect at biface, frontal orbit, frontotemporal, and random position, respectively. **a** The defective cranium; **b** the ground truth implant; **c** the automatically generated implant; **d** distance map viewed from the outside; **e** distance map viewed from the inside



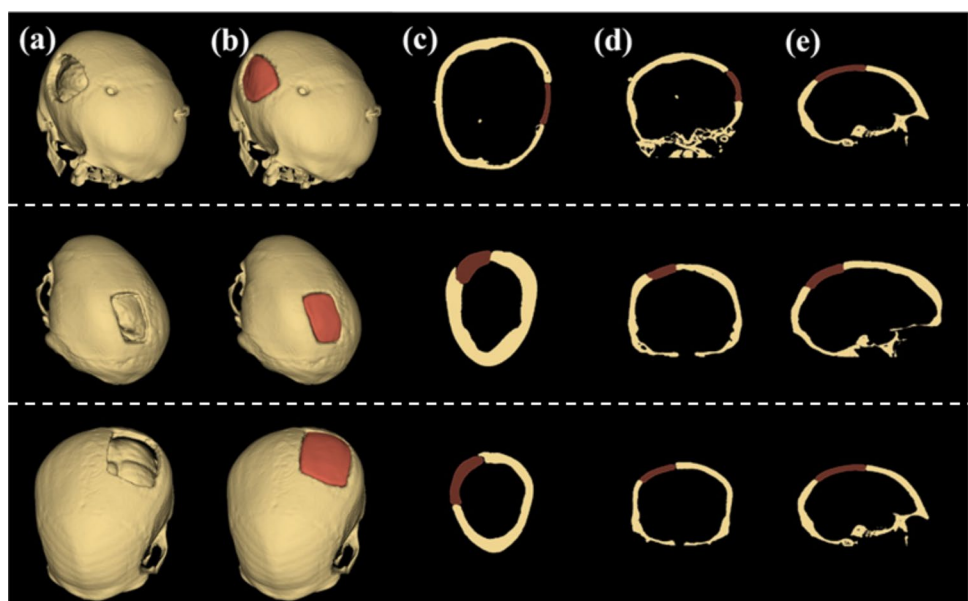
absolute maximum distance of the overly concave regions is usually lower than that of the overly convex regions. Thus, the repaired cranium may be a little thicker on the defective region than its original shape.

The automatically generated implant (c) in the first row hits the top of the bounding box, which causes irregular flatland. The fact that the bounding box of defective cranium may be smaller than that of repaired cranium, and inappropriate clipping of voxel images are both associated with this phenomenon.

5.1 Real craniotomy cases

Craniotomy is a surgical procedure where a part of the skull is removed to access the brain for treatment. It involves opening a section of the skull and using a surgical tool to remove a portion of the skull bone to reach brain. The performance of the proposed workflow is also evaluated on the 11 real defective crania. Since there are no ground-truth implants provided for these cases, evaluation can only be conducted visually and subjectively. The proposed workflow

Fig. 8 Graphical views of some real craniotomy cases with automatically generated cranial implants. **a** The defective cranium; **b** the defective cranium with the corresponding implant; **c–e** slice views of the defective cranium with the corresponding implant of axial, coronal, and sagittal planes, respectively



can generate acceptable cranial implants for most of these cases, except three cases, for which the outputs are either incomplete or irregular. These successful real craniectomy cases are randomly chosen and demonstrated in Fig. 8a the defected crania, Figure 8b the defective crania with corresponding implant, and Fig. 8c–e slice views of the defective crania with the corresponding implant of axial, coronal, and sagittal plane respectively. For these cases, the generated implants can visually fill the gap between the defective edges in a smooth way, which also shows the generalization ability of the proposed workflow from synthetic cases to real cases.

6 Discussion

Although the proposed workflow works well on the synthesized dataset and eight out of 11 real craniotomy cases, but still failed to work on the rest three cases. There are mainly two types of flaw regions on the failed implants, which are hyperplasia and hollow. The three cases shown in Fig. 9 are named as a, b, and c failure case, respectively.

Hyperplasia regions appear in the a and c failure cases, and hollow regions appear in the b and c failure cases. The hyperplasia regions exist on the boundary of the implants and the skull thickness of the cases with hollow regions is visually thinner than others.

1. Quantitative analysis of failed cases:

Skull Thickness Measurement: We have performed precise measurements of skull thickness in the regions surrounding the hollow defects. Specifically, we measured the skull thickness at multiple points within and

adjacent to the hollow regions in cases (b) and (c), comparing these measurements to the average skull thickness in successful cases. Our analysis revealed that the average skull thickness within the hollow regions was significantly lower ($4.2 \text{ mm} \pm 0.8 \text{ mm}$) compared to the adjacent areas ($6.1 \text{ mm} \pm 1.2 \text{ mm}$) ($p < 0.05$, t-test), supporting our hypothesis that thinner skull thickness contributes to hollow defect formation.

Hyperplasia Distribution Analysis: We have analyzed the distribution of hyperplasia along the implant boundaries in cases (a) and (c). We measured the length of the hyperplasia regions as a percentage of the total implant boundary. We also analyzed the curvature of the skull boundary in these regions.

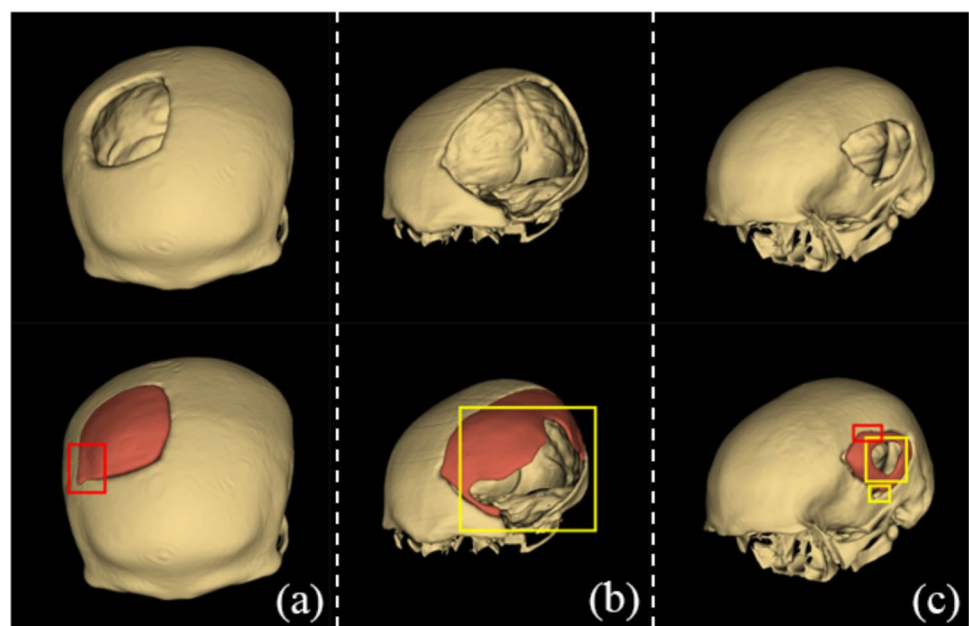
Failure Distribution Analysis: We have also analyzed the distribution of the prediction failures along the edges of the defects. For case (b), we have now quantified the proportion of the upper and lower defect regions that were successfully filled. This reinforces our observation regarding the influence of defect position and potentially suggests a directional bias in the current model.

2. Potential solutions to minimize failures:

Based on the quantitative analysis, we have explored and implemented several potential solutions to minimize these failures:

Addressing Hyperplasia: To address hyperplasia, we have augmented our training dataset with additional examples of complex skull boundaries and synthetic data augmentation techniques to improve the model's ability to generalize to unseen boundary shapes. This includes adding more training examples with varied curvature and boundary definitions.

Fig. 9 Graphical view of the three failed real craniectomy cases. Hyperplasia regions appear in the (a) and (c) failure cases and hollow regions appear in the (b) and (c) failure cases. Red and yellow boxes indicate the outstanding hyperplasia and hollow regions, respectively



Mitigating Hollow Regions: To mitigate hollow regions, we are exploring incorporating local skull thickness information directly into the implant design process. We are investigating:

using a pre-trained model to estimate bone thickness from CT data and incorporating this into the implant design optimization.

Improving Defect Position Handling: To improve the handling of defect positions, we are investigating the use of attention mechanisms in our model. We hypothesize that these mechanisms can help the model focus on the relevant features of the defect region and improve the consistency of predictions across different locations within the defect.

During the post-processing stage, upsampling the predicted implant to the original CT scan resolution is necessary. As is common with upsampling, this process can introduce minor jaggedness or discontinuities in the implant surface. While our primary focus in this work is on demonstrating the feasibility and accuracy of the automated design process, we acknowledge that these surface irregularities could be further addressed in future work. We have employed a Gaussian blur as a final smoothing step to mitigate these visual artifacts. A more detailed investigation of optimal upsampling and smoothing techniques, including exploring interpolation methods prior to upsampling and potential network architecture refinements to minimize these irregularities at the source, will be a focus of future research.

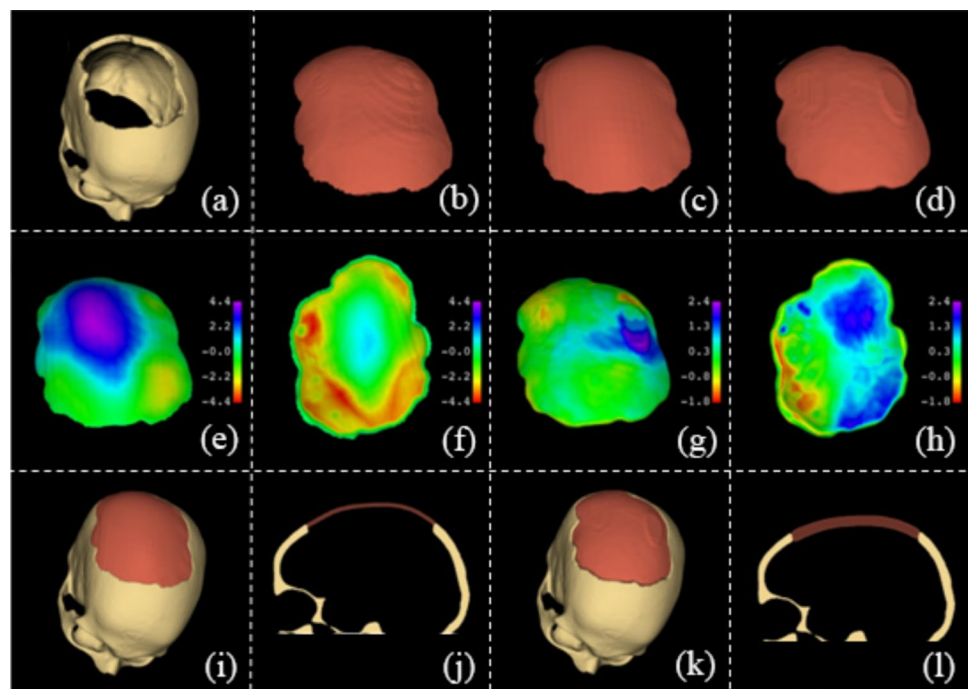
6.1 Comparison with manually designed cranial implants

In addition to the proposed automatic design workflow, we conducted a comparative analysis with a manual design process using Geomagic Sculpt (3D Systems, USA), a 3D design software designed for the creation and editing of 3D meshes [22]. The manual design process involved the following steps:

1. Select various points along the defect region from the outside surface of the defective cranium model.
2. Use the “Fill” function to close the selected region and form a surface.
3. Use the “Subdivision” function to add extra points uniformly on the implant surface.
4. Use the “Curve Network” tool to manually adjust the shape of the implant surface.
5. Use the “Extrude” function to form the implant of a given thickness.
6. Use Boolean operation to refine the implant.

We selected one case from the datasets and designed the implant manually and automatically respectively. As shown in Fig. 10, the AI-based design result was better than the manual method. Specifically, in subfigures e–h, the color legends indicate that the automatically generated implant has a surface distance error that is closer to the ground truth compared to the manually designed implant. Moreover, on the external side of the manually designed implant, there

Fig. 10 Graphical comparison of the manually designed and the automatically generated implants. **a** The defective cranium; **b** the ground truth implant; **c** the manually designed implant; **d** the automatically generated implant; **e–f** distance map of the manually designed implant with the ground truth, viewed from outside and inside, respectively; **g–h** distance map of the automatically generated implant with the ground truth, viewed from outside and inside, respectively; **i–j** 3D and slice view of the repaired cranium with manually designed implant; **k–l** 3D and slice view of the repaired cranium with automatically generated implant



is a large centralized overly convex region, while the same region in the automatically generated implant appears to be more similar to the ground truth, as illustrated in sub-figures e and f.

In contrast to the manually designed cranial implant, which is limited to having a unique thickness due to the nature of the manual design process, the proposed automatic implant design workflow does not have this limitation. Additionally, the proposed automatic workflow can replace all the tedious steps with just a few clicks, with the aid of the 3D Slicer plugin. This feature can effectively reduce the workload of the cranial implant designers.

Overall, the results demonstrate that the proposed automatic implant design workflow is superior to the manual design process in terms of accuracy and efficiency, making it a promising solution for cranial implant design.

7 Future work

At present, the absence of data augmentation methods during the training of the deep neural network leads to the occurrence of overfitting, wherein the loss value on the training set is substantially lower than that on the testing set. As such, it is crucial to implement appropriate data augmentation strategies to counteract overfitting and improve the generalization ability of the model. We will investigate various data augmentation strategies to mitigate overfitting, such as random rotations, scaling, mirroring, elastic deformations, and noise addition. We will also explore more advanced techniques like adversarial training.

Future work will also include a more rigorous quantitative evaluation of the proposed method, including the calculation of mean absolute error (MAE) and other relevant metrics to compare the automated design process with manual methods. This will provide a more objective assessment of the accuracy and reliability of the proposed workflow.

Moreover, in order to validate the applicability of the proposed workflow in real-world clinical scenarios, further experiments need to be conducted. To this end, we intend to create more realistic defective cases using healthy crania by simulating defects using 3D modeling software. We will then test the workflow on these simulated and real craniectomy cases, including cases with complex defect shapes and defects near critical anatomical structures. This will allow us to assess the robustness and generalizability of the proposed method in real-world clinical settings. Furthermore, future work will address the current limitations of the method in handling very large defects. We will also explore ways to improve the computational efficiency of the workflow to make it more suitable for clinical use.

Future work will also address the important ethical considerations surrounding the clinical application of automated cranial implant design. A crucial aspect is establishing clear lines of responsibility and accountability in the clinical workflow, particularly in cases of errors or adverse outcomes. While the AI can assist in the design process, ultimate responsibility for patient safety must reside with qualified medical professionals. We will also prioritize mitigating potential biases in the AI system by expanding the training dataset to include a more diverse and representative patient population. Furthermore, we will investigate methods to improve the transparency and explainability of our deep learning model, allowing clinicians to understand the rationale behind the generated implant designs. Human oversight protocols will be developed to ensure that surgeons have the necessary tools and information to review and customize the automated designs before implantation. Finally, we will adhere to strict data privacy and security protocols, complying with all relevant regulations and ethical guidelines, to protect patient data throughout the process. Future studies will also focus on developing robust informed consent procedures that clearly explain the use of AI in cranial implant design to patients, ensuring they understand the potential benefits and risks.

8 Conclusion

This study presents an innovative automatic cranial implant design workflow utilizing a deep neural network to predict the original shape of a defective region of a cranium. The proposed workflow's shape completion ability was evaluated using cross-validation, resulting in DSC, bDSC, and 95hd measurements of 0.81, 0.81, and 3.00 mm, respectively. The visually satisfactory results achieved in the comparison with the manual design process suggest the potential clinical applications of the proposed workflow.

While this study demonstrates the feasibility and potential of the automated method, a more rigorous quantitative comparison with manual designs, including metrics such as mean absolute error (MAE) and standard deviation, remains an important direction for future work. Future studies will focus on addressing this limitation and further validating the clinical utility of the proposed workflow.

Funding This work was supported by grants from the National Natural Science Foundation of China (82330063), Shanghai Jiao Tong University Foundation on Medical and Technological Joint Science Research (YG2023ZD19; YG2023ZD15; YG2024QNA23; YG2024QNB26), the Foundation of Science and Technology Commission of Shanghai Municipality (24TS1413000, 24490710300,

22Y11911700), and Shanghai Leading Talent Program of Eastern Talent Plan(BJKJ2024003).

Declarations

Competing interests The authors declare no competing interests.

References

1. Van GJ, Forouzanfar T, Liberton NP et al (2018) Accuracy of computer assisted surgery in mandibular reconstruction: a systematic review. *Oral Oncol* 84:52–60
2. Zhao L, Patel PK, Cohen M (2012) Application of virtual surgical planning with computer assisted design and manufacturing technology to cranio-maxillofacial surgery. *Arch of Plast Surg* 39(4):309
3. Li J, Campe G, Pepe A et al (2021) Automatic skull defect restoration and cranial implant generation for cranioplasty. *Med Image Anal* 73(102171):1–15
4. Moiduddin K, Darwish S, Al-Ahmari A, ElWatidy S, Mohammad A, Ameen W (2017) Structural and mechanical characterization of custom design cranial implant created using additive manufacturing. *Electron J Biotechnol* 1(29):22–31
5. Vallittu PK (2017) Bioactive glass-containing cranial implants: an overview. *Journal of Materials Science*. 52(15):8772–84
6. Campe GV, Pistracher K (2020) Patient specific implants (PSI). *Cranioplasty in Neurosurgical Clinical Routine*. Springer, pp 1–9
7. Li J, Pepe A, Gsaxner C, Egger J (2020) An online platform for automatic skull defect restoration and cranial implant design. <https://doi.org/10.48550/arXiv.2006.00980>
8. Chen X, Xu L, Li X, Egger J (2017) Computer-aided implant design for the restoration of cranial defects. *Sci Rep* 7(1):1–0
9. Egger J, Gall M, Tax A, Ücal M, Zefferer U, Li X, von Campe G, Schäfer U, Schmalstieg D, Chen X (2017Mar 6) Interactive reconstructions of cranial 3D implants under MeVisLab as an alternative to commercial planning software. *PLoS One* 12(3):e0172694
10. Wu CT, Yang YH, Chang YZ (2022) Three-dimensional deep learning to automatically generate cranial implant geometry. *Sci Rep* 12(1):1
11. Mainprize JG, Fishman Z, Hardisty MR (2020) Shape completion by U-Net: an approach to the Auto-Implant MICCAI cranial implant design challenge. In: *Cranial implant design challenge*. Springer, Cham, pp 65–76
12. Bayat A, Shit S, Kilian A, Liechtenstein JT, Kirschke JS, Menze BH (2020) Cranial implant prediction using low-resolution 3D shape completion and high-resolution 2D refinement. In: *Cranial Implant Design Challenge*. Springer, pp 77–84
13. Kikinis R, Pieper SD, Vosburgh KG (2014) 3D Slicer: a platform for subject-specific image analysis, visualization, and clinical support. In: *Intraoperative imaging and image-guided therapy*. Springer, pp 277–289
14. Ronneberger O, Fischer P, Brox T (2015) U-net: convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention. MICCAI 2015. Lecture Notes in Computer Science* 9351:234–241. https://doi.org/10.1007/978-3-319-24574-4_28
15. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*, pp 770–778. <https://doi.org/10.1109/CVPR.2016.90>
16. Xu B, Wang N, Chen T, Li M (2015) Empirical evaluation of rectified activations in convolutional network. <https://arxiv.org/abs/1505.00853>
17. Li J, Pimentel P, Szengel A, Ehlke M, Lamecker H, Zachow S, Estacio L, Doenitz C, Ramm H, Shi H, Chen X (2021May 3) AutoImplant 2020-first MICCAI challenge on automatic cranial implant design. *IEEE Trans Med Imaging* 40(9):2329–42
18. Kodym O, Li J, Pepe A, Gsaxner C, Chilamkurthy S, Egger J, Španěl M (2021Apr) SkullBreak/SkullFix–Dataset for automatic cranial implant design and a benchmark for volumetric shape learning tasks. *Data in Brief*. 1(35):106902
19. Dice LR (1945) Measures of the amount of ecologic association between species. *Ecology* 26(3):297–302
20. Crum WR, Camara O, Hill DL (2006) Generalized overlap measures for evaluation and validation in medical image analysis. *IEEE Trans Med Imaging* 25(11):1451–1461
21. Cignoni P, Rocchini C, Scopigno R (1998) Metro: measuring error on simplified surfaces. *Computer Graphics Forum* 17(2):167–174
22. Van der Meer WJ, Bos RR, Vissink A, Visser A (2013) Digital planning of cranial implants. *Br J Oral Maxillofac Surg* 51(5):450–2

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Afaq Rafique Memon received the PhD degree in Mechanical Engineering from Shanghai Jiao Tong University, China, in 2023. He is currently working as Workshop Instructor in Mechanical Engineering Workshop, Mehran University of Engineering and Technology, Jamshoro, Pakistan. His research focuses on computer-assisted surgery, including medical image analysis, computer graphics, implant design, medical 3D printing, and CAD/CAM for mechanical equipment.



Haochen Shi received the BE degree from Shanghai Jiao Tong University China, in 2022. He is currently working toward the PhD's degree with the School of Mechanical Engineering, Shanghai Jiao Tong University. His supervisor is Prof. Xiaojun Chen. His research interests include computer graphics and image processing.



Dr. Tarique Rafique Memon earned his Ph.D. in 2024 from Mehran University of Engineering and Technology, Jamshoro, Sindh, Pakistan, with a focus on image processing and instrumentation systems. He has been an integral part of the Department of Electronic Engineering at Quaid-e-Awam University of Engineering, Science, and Technology (QUEST), Nawabshah, since 2006, where he serves as an Assistant Professor. His primary areas of expertise include embedded systems,

instrumentation systems, and IoT devices, where he has conducted extensive research and development.



Jan Egger received his Diploma, Master, and PhD degrees in computer science in 2004, 2006 and 2009, respectively. He also has PhD degree in human biology (2012) and a Habilitation in computer science (2016). Currently, he is a professor and leads the AI-guided therapies (AIT) group at the Institute for Artificial Intelligence in Medicine (IKIM) and is deputy chairperson of the Center for Virtual and Extended Reality in Medicine (ZvRM). His main research interests are medical image processing and extended reality.



Xiaojun Chen received the PhD degree in mechanical engineering from Shanghai Jiao Tong University, China, in 2006. He is currently a professor of the School of Mechanical Engineering, Shanghai Jiao Tong University. His research focuses on computer-assisted surgery, including medical image analysis, computer graphics, surgical navigation, VR/AR technology in medicine, surgical robotics, and medical 3D printing.